

Day 11 - Notes

DAY 11 – Testing, Asynchronous Processing & Enterprise Reliability

Apex Testing

Testing is used to verify that enterprise applications work correctly under different conditions.

Why Testing Matters

- Prevents production bugs
 - Improves reliability
 - Ensures business logic works correctly
 - Maintains data consistency
 - Helps safe deployment of code
 - Reduces application failures
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Unit Testing

Unit testing verifies individual methods and functionalities.

Examples:

- Student registration validation
 - Attendance calculations
 - Fee payment updates
 - Login authentication
 - Notification generation
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Important Test Cases for College Management System

1. Invalid Email Format

Prevents incorrect student records.

2. Duplicate Registration

Prevents multiple registrations for same student.

3. Seats Exceeding Limit

Ensures admission capacity is not exceeded.

4. Attendance Below Threshold

Prevents ineligible students from exams.

5. Payment Failure

Avoids incomplete financial records.

6. Invalid Login Credentials

Protects system security.

7. Empty Mandatory Fields

Ensures required data is entered.

8. Incorrect Marks Entry

Prevents wrong grade calculations.

9. Notification Failure

Ensures alerts are properly monitored.

10. Simultaneous Registration Requests

Tests system stability during high traffic.

Asynchronous Processing

Asynchronous processing allows operations to run in the background without blocking users.

Benefits

- Better performance
- Improved scalability
- Faster user experience
- Efficient processing

Types of Async Apex

Future Methods

Used for lightweight background tasks.

Queueable Apex

Supports job chaining and monitoring.

Batch Apex

Processes large amounts of records in batches.

Scheduled Apex

Runs jobs automatically at scheduled times.

Synchronous vs Asynchronous Execution

Synchronous Processing

- Runs immediately
 - User waits until completion
 - Suitable for small tasks
- Example:
Saving a single student record

Asynchronous Processing

- Runs in background
 - User continues working
 - Suitable for heavy processing
- Example:
Bulk email sending
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Async Processing Use Cases

1. Bulk Emails

Sending notifications to thousands of students.

2. Report Generation

Generating large reports in background.

3. Large Data Import

Processing huge datasets efficiently.

4. External System Synchronization

Syncing payment or external portal data.

5. Notification Processing

Sending SMS or alerts asynchronously.

Reliability Thinking

Enterprise systems must continue working correctly even during failures.

Possible Problems During Failure

Student Registration Failure

- Duplicate records
- Missing data
- Partial registration

Payment Update Failure

- Incorrect fee status
- Duplicate payments
- Financial inconsistencies

Attendance Update Failure

- Wrong attendance percentage
- Incorrect eligibility calculations

How Testing Helps

- Detects bugs early
 - Validates workflows
 - Improves system stability
 - Ensures reliable transactions
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Scalability

Scalability means the system can handle increasing users and data without performance problems.

Enterprise systems must support:

- Thousands of users
 - Large databases
 - Continuous operations
 - Heavy workloads
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Why Enterprise Systems Use Background Jobs

- Avoid blocking operations
 - Improve responsiveness
 - Handle large processing efficiently
 - Maintain smooth user experience
 - Process heavy workloads reliably
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Key Concepts Learned

- Testing improves reliability
 - Unit tests prevent failures
 - Async processing improves performance
 - Future, Queueable, Batch, and Scheduled Apex support background processing
 - Enterprise systems require scalability
 - Reliability engineering is important
 - Large systems must avoid blocking operations
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Reflection

Enterprise systems require testing, scalability, and asynchronous processing because they handle large amounts of users, records, and business operations simultaneously. Direct execution alone is not sufficient for enterprise applications because heavy operations can slow down or crash systems.

Testing ensures reliability and prevents failures. Scalability helps systems support growth efficiently. Asynchronous processing allows resource-intensive operations to run in the background while maintaining smooth application performance and better user experience.

Enterprise software differs from small scripts because it must operate continuously, reliably, securely, and efficiently at large scale.

Final Outcome

Understood enterprise reliability, testing strategies, scalability concepts, and asynchronous processing in Salesforce. Learned how enterprise systems use testing and background jobs to build stable, scalable, and production-ready applications.